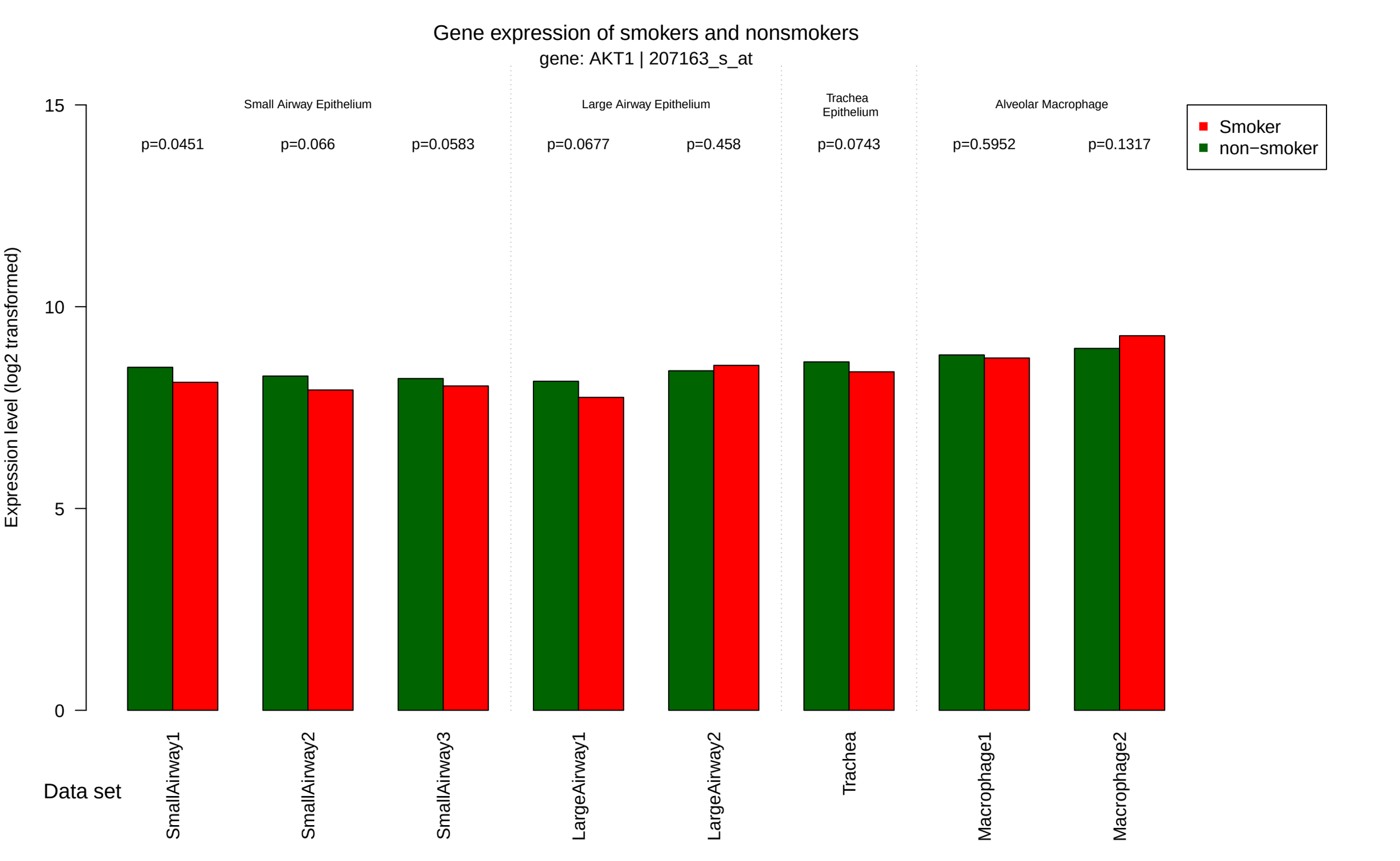
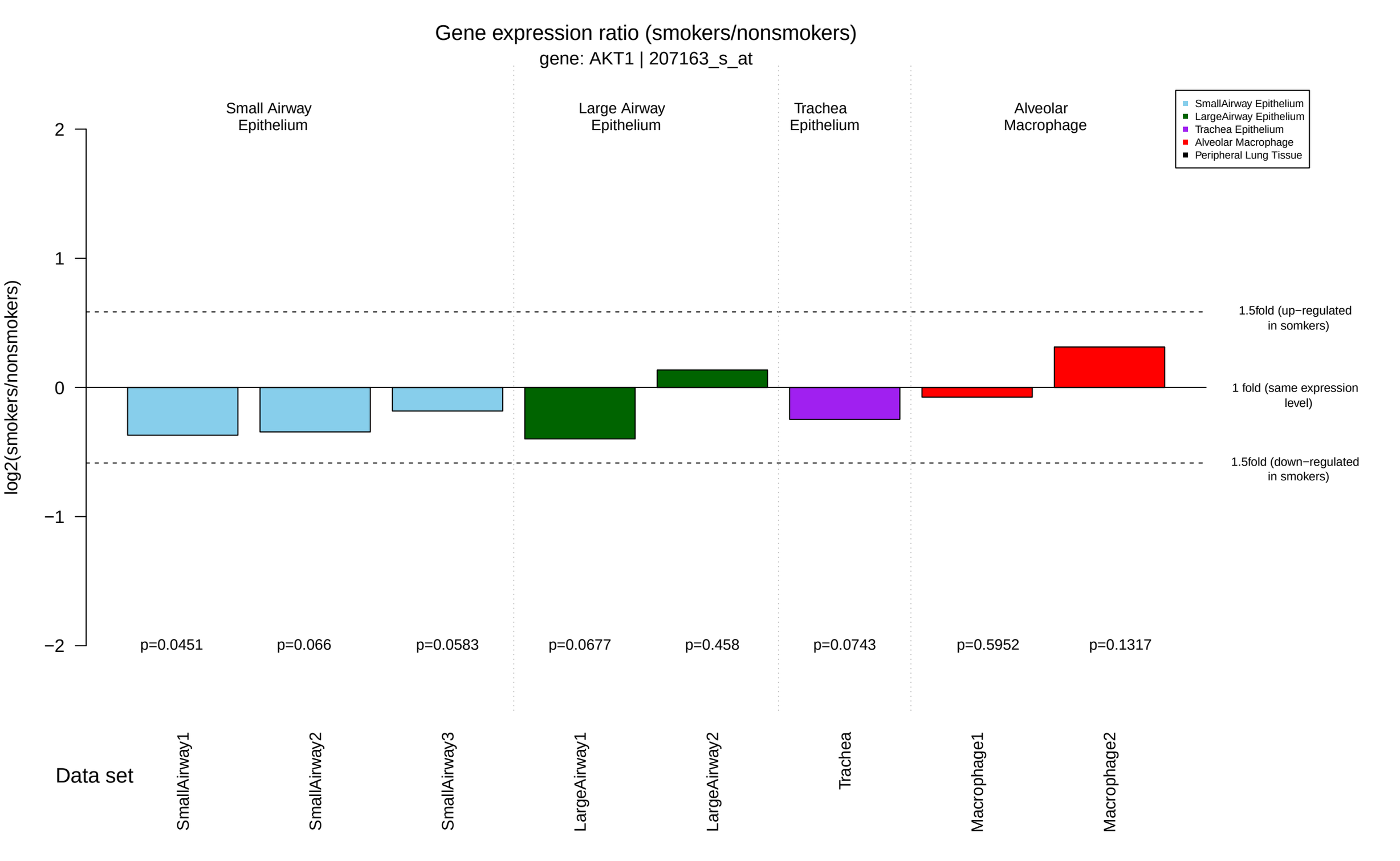
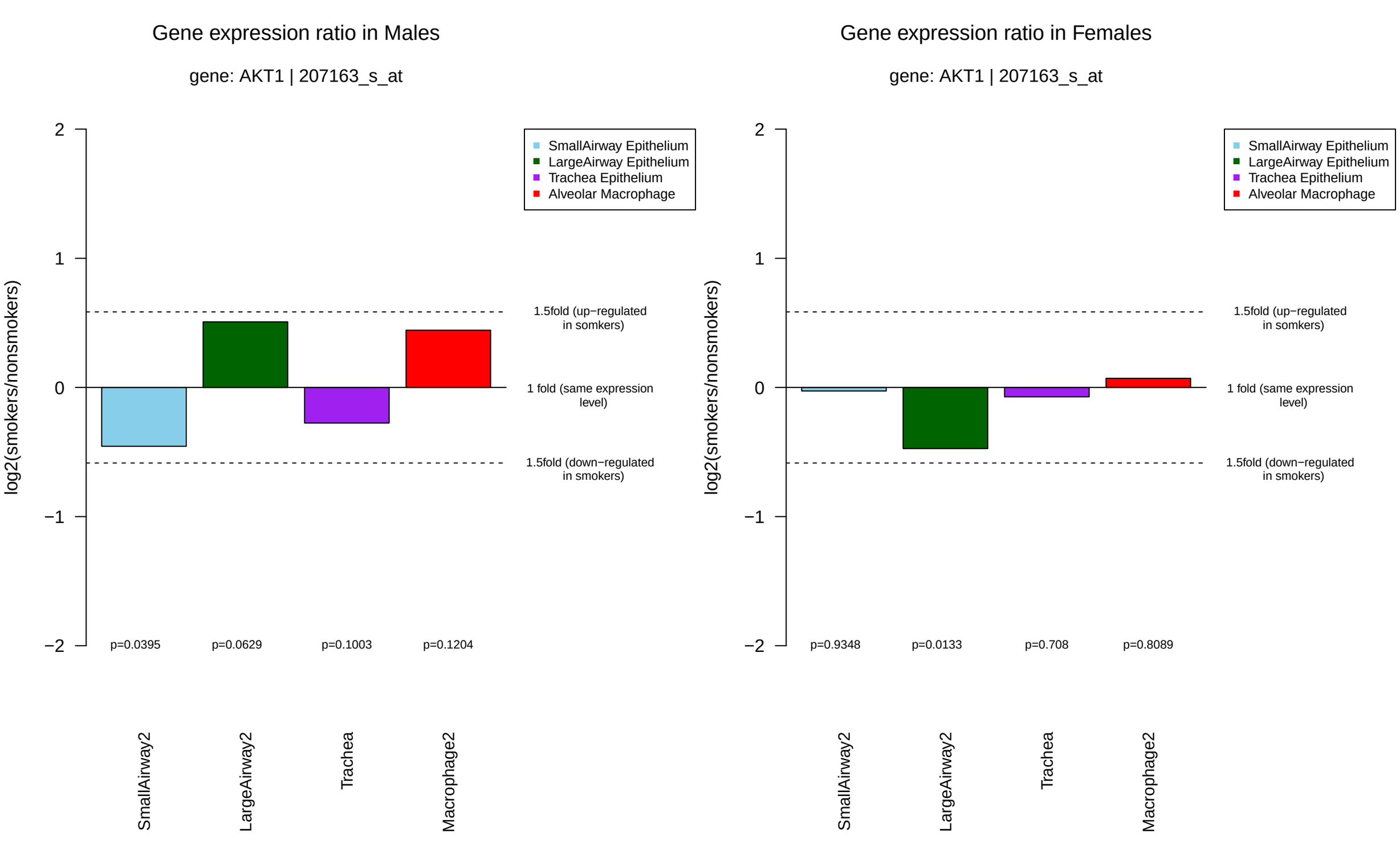
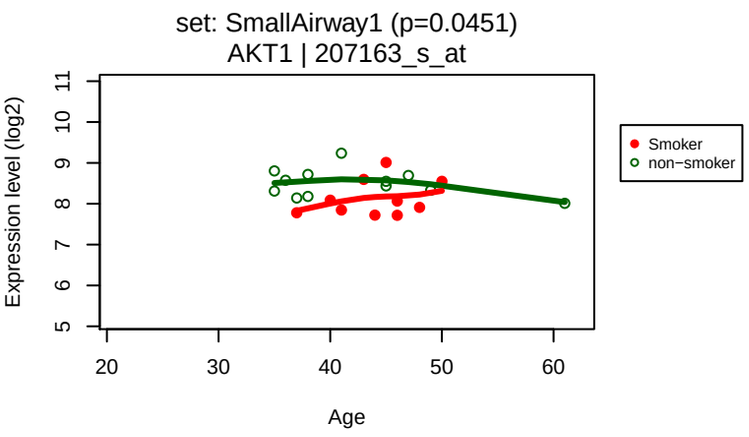
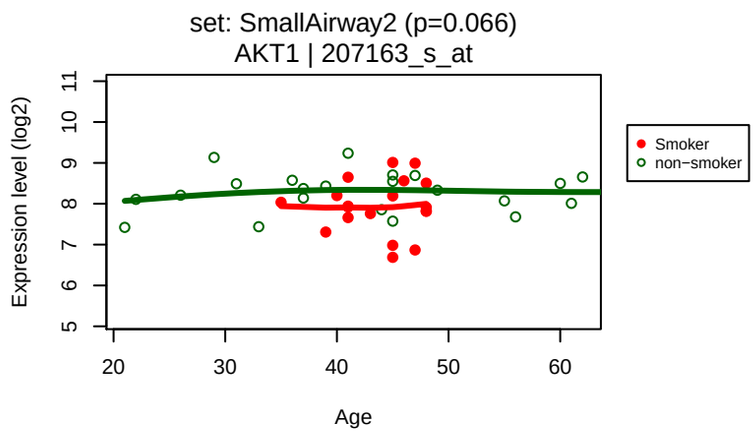
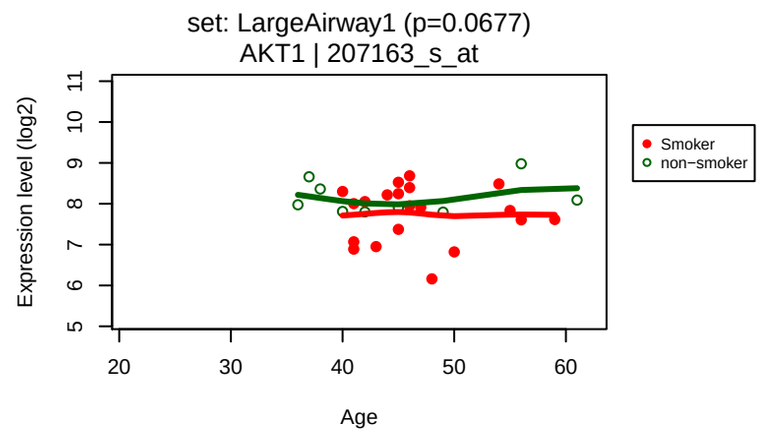

Gene expression of smokers and nonsmokers
gene: AKT1 | 207163_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0451
p=0.066
p=0.0583
p=0.0677
p=0.458
p=0.0743
p=0.5952
p=0.1317
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

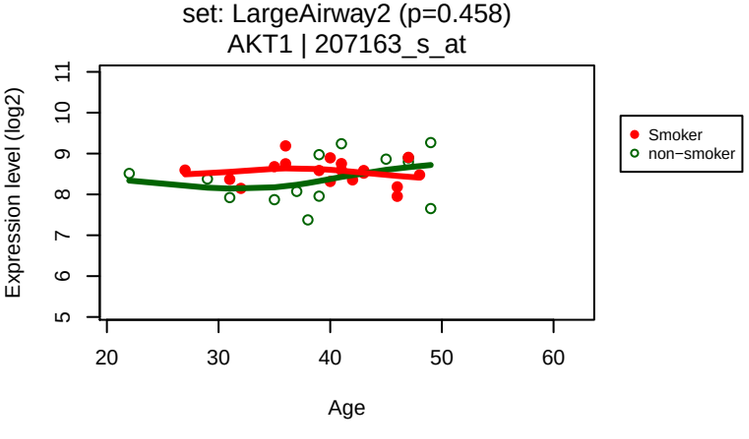

Gene expression ratio (smokers/nonsmokers)
gene: AKT1 | 207163_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0451
p=0.066
p=0.0583
p=0.0677
p=0.458
p=0.0743
p=0.5952
p=0.1317
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

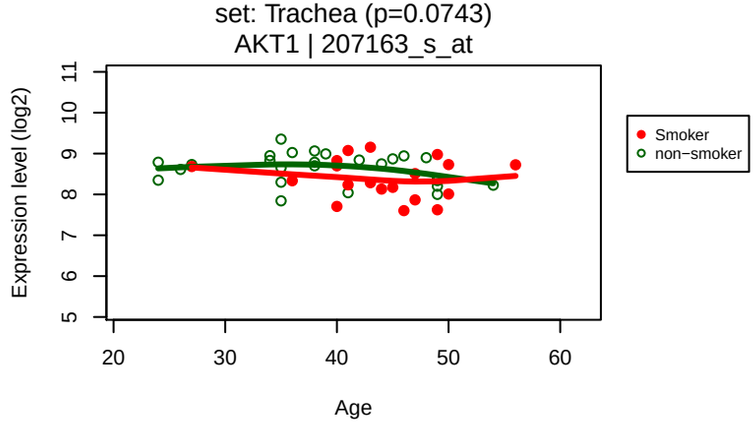

Gene expression ratio in Males
gene: AKT1 | 207163_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0395
p=0.0629
p=0.1003
p=0.1204
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT1 | 207163_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9348
p=0.0133
p=0.708
p=0.8089
SmallAirway2
LargeAirway2
Trachea
Macrophage2

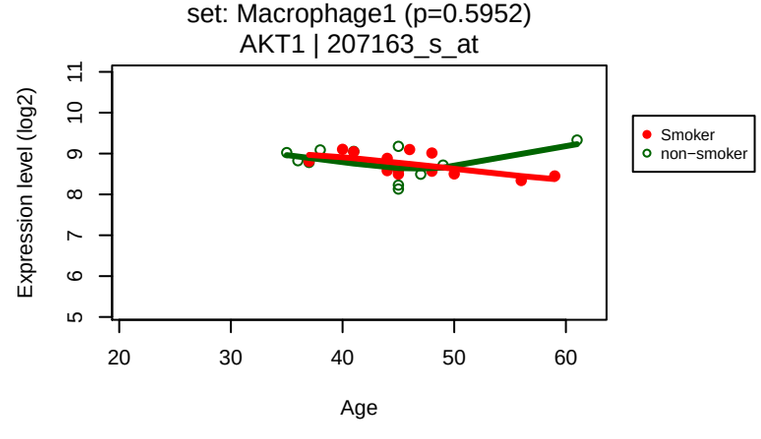

set: SmallAirway1 (p=0.0451)
AKT1 | 207163_s_at
Expression level (log2)
Age
Smoker
non−smoker

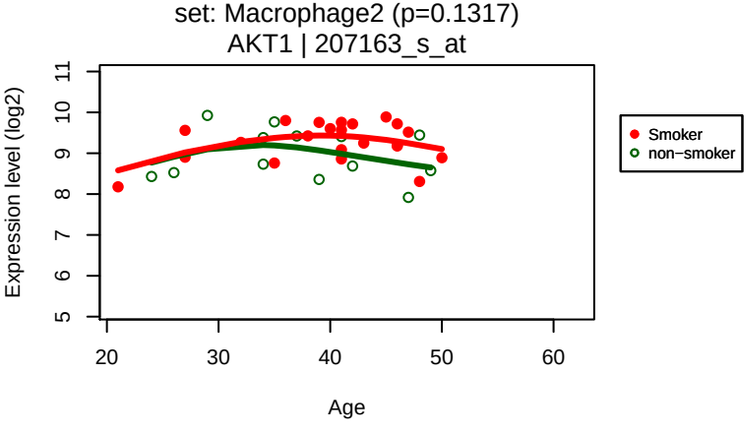

set: SmallAirway2 (p=0.066)
AKT1 | 207163_s_at
Expression level (log2)
Age
Smoker
non−smoker

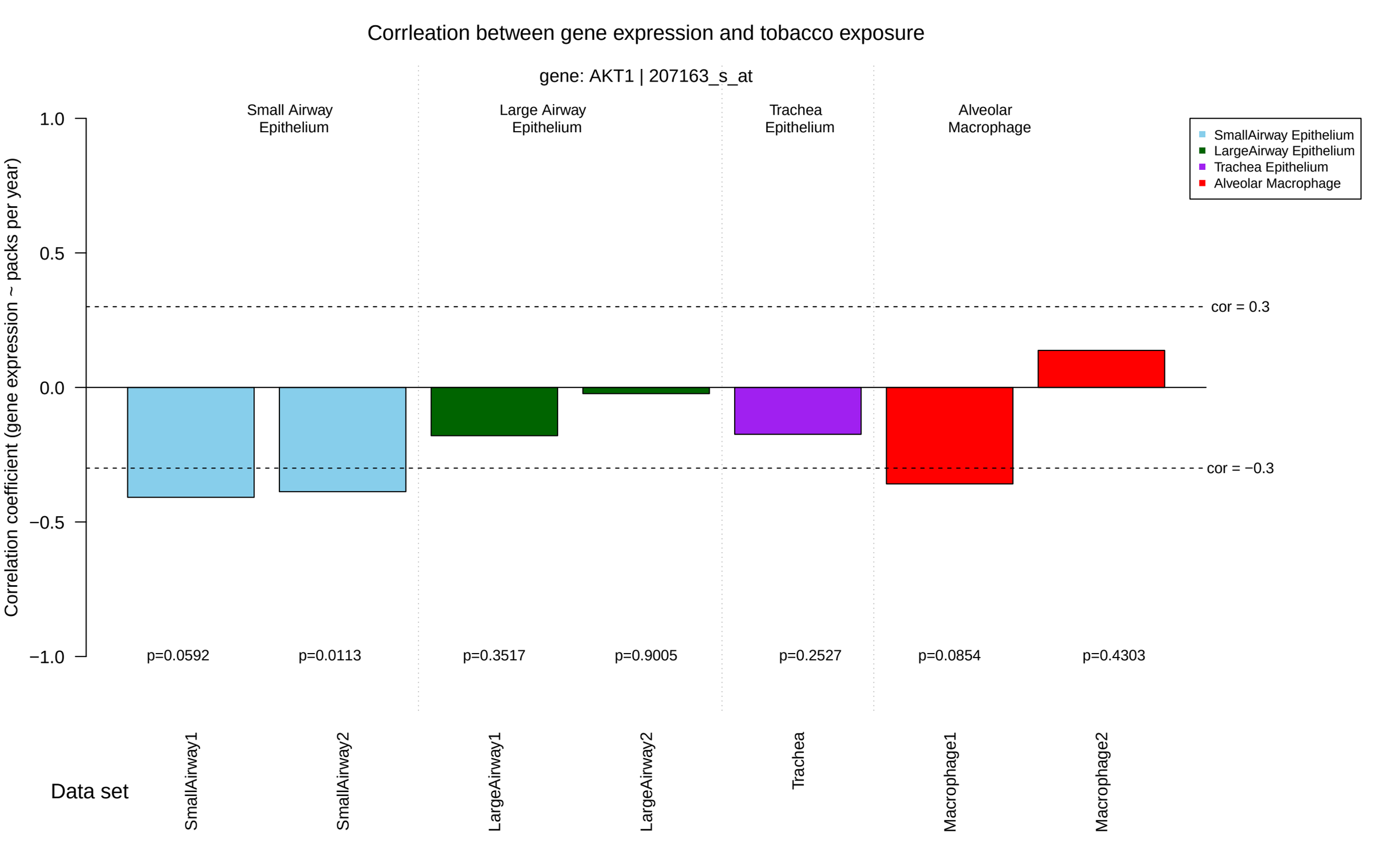

set: LargeAirway1 (p=0.0677)
AKT1 | 207163_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.458)
AKT1 | 207163_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0743)
AKT1 | 207163_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5952)
AKT1 | 207163_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1317)
AKT1 | 207163_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT1 | 207163_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0592
p=0.0113
p=0.3517
p=0.9005
p=0.2527
p=0.0854
p=0.4303
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2